MATE664 L07
Jan-26 2026

Recap: One equation to rule them all:
Fick's second law (General time-dependent equation)

$$\frac{\partial c}{\partial t} = -\nabla\cdot[D\nabla c] + \cancel{Gen}\ 0$$

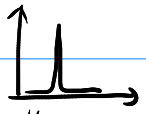
D isotropic? N → Variant D system

Y

$\frac{\partial c}{\partial t} = 0$? → Steady state solution

- Integration
- Coordinate (∇?)
- B.C.

$\frac{\partial c}{\partial t} \neq 0$? → Unsteady state / Time-dependent

1. semi-half space solution
    line solution
   ⇓ Superposition method
2. separation of variables / Fourier transform
3. Laplace transform
4. Numerical integration

Recap II.

Steady state solution

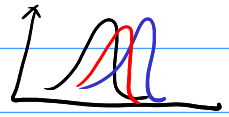
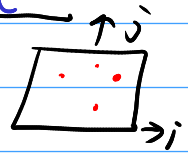
We need to distinguish geometries $\nabla \rightrightarrows$

- Cartesian
- Cylindrical
- Spherical

$\xrightarrow{\text{See demo}}$ for details

Part II Non-steady state

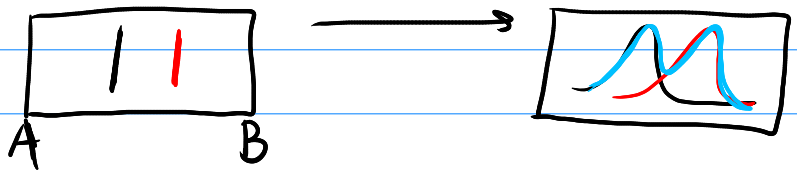
Different strategies exist

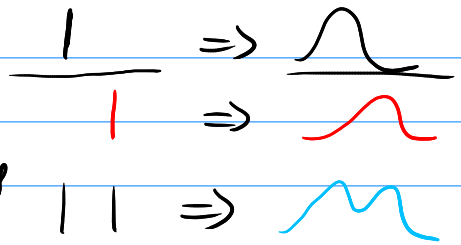
- Superposition + well-known source solutions 
- Laplace transform (real PDE → imaginary s-plane coeffs) 
- Numerical solution

Method 1

① Many cases use a well-know geometry

② If a geometry is complex ← addition of multiple terms (superposition)



Initial cond  ⇒ Solution a

⇒ Solution b

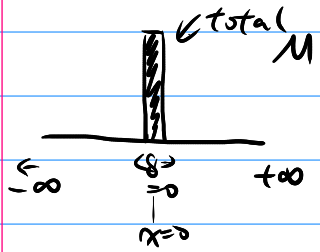
Combined source ⇒ Solution = a+b

Boundary conditions additive

In actual lecture talked later

Infinitely large system $\frac{dc}{dt} = D\frac{d^2c}{dx^2}$ ← Dirac delta func

$$C(x,0) = N\delta(x)$$



Solution

$$C(x,t) = \frac{K}{\sqrt{t}}\exp\left(-\frac{x^2}{4Dt}\right)$$

↑ Gaussian

Conservation $N = \int_{-\infty}^{+\infty} C(x)\,dx = \int_{-\infty}^{\infty}\frac{A}{\sqrt{t}}\exp\left(-\frac{x^2}{4Dt}\right)$

$$A = \frac{M}{2\sqrt{\pi D}} \qquad C(x,t) = \frac{N}{2\sqrt{\pi D t}}\exp\left(-\frac{x^2}{4Dt}\right)$$

How do we get there ? (Bonus part)

we have $\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$ $\begin{cases} \text{LHS - time} \\ \text{RHS - Coord} \end{cases}$

Recall $\sqrt{Dt}$ has length unit $\Rightarrow$ dimensionless transform

$$\eta = \frac{x}{\sqrt{Dt}} \Rightarrow \frac{\partial c}{\partial t} = \frac{\partial c}{\partial \eta}\cdot\frac{\partial \eta}{\partial t} = \frac{\partial c}{\partial \eta}\cdot\frac{x}{\sqrt{D}}\cdot\frac{1}{\sqrt{t^3}}\cdot -\frac{1}{2}$$

$t^{-\frac{1}{2}}$

$$= -\frac{1}{2}\frac{x}{\sqrt{Dt}}\cdot\frac{1}{t}\frac{\partial c}{\partial \eta}$$

$$= -\frac{1}{2}\eta\cdot\frac{1}{t}\cdot\frac{\partial c}{\partial \eta}$$

$$\frac{\partial^2 c}{\partial x^2} = \frac{\partial^2 c}{\partial(\eta\cdot\sqrt{Dt})^2} = \frac{1}{Dt}\frac{\partial^2 c}{\partial \eta^2}$$

$$\Rightarrow \frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} \Rightarrow -\frac{1}{2}\eta\frac{1}{\cancel{t}}\frac{\partial c}{\partial \eta} = \cancel{D}\cdot\frac{1}{\cancel{D}\cancel{t}}\frac{\partial^2 c}{\partial \eta^2}$$

$$\frac{\partial^2 c}{\partial \eta^2} + \frac{1}{2}\eta\frac{\partial c}{\partial \eta} = 0 \qquad \text{Let } u = \frac{\partial c}{\partial \eta}$$

$$\frac{\partial u}{\partial \eta} + \frac{1}{2}\eta u = 0 \qquad \frac{\partial u}{u} = -\frac{1}{2}\eta\,\partial\eta$$

$$\int\frac{du}{u} = \int -\frac{1}{2}\eta\,d\eta \Rightarrow \ln u = -\frac{1}{4}\eta^2 + \text{Const}$$

$$u = C_1\exp(-\tfrac{1}{4}\eta^2)$$

since $u = \frac{\partial c}{\partial \eta}$

$$\frac{\partial c}{\partial \eta} = C_1\exp(-\tfrac{1}{4}\eta^2) \Rightarrow \int dc = \int C_1\exp(-\tfrac{1}{4}\eta^2)\,d\eta$$

$$c(\eta) = \overset{\text{coeff}}{\square} + \overset{\text{coeff}}{\square}\underbrace{\int\exp(-\tfrac{1}{4}\eta^2)\,d\eta}$$

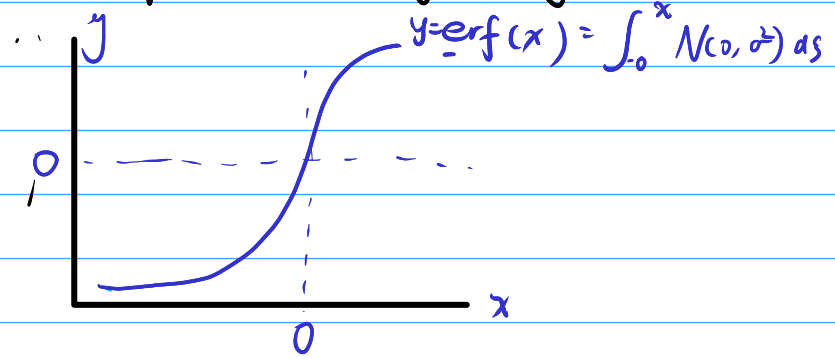
error function form

$$\operatorname{erf}(\xi) = \frac{2}{\sqrt{\pi}}\int_0^{\xi} e^{-x^2}\,dx$$

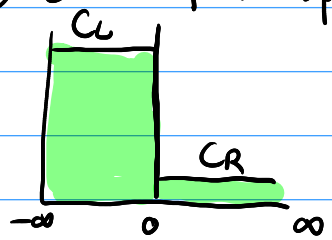
$$= k_1 + k_2\operatorname{erf}\left(\frac{\eta}{2}\right) = \boxed{k_1 + k_2\operatorname{erf}\left(\frac{x}{\sqrt{4Dt}}\right)}$$

k_1 & k_2 are determined by the initial & B.C

Error function: integral of gaussian

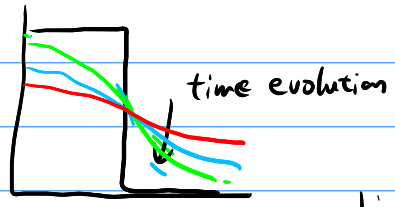


① Semi-infinite space



$$C(\eta) = \frac{C_L + C_R}{2} + \frac{C_R - C_L}{2}\,\mathrm{erf}\left(\frac{\eta}{2}\right)$$

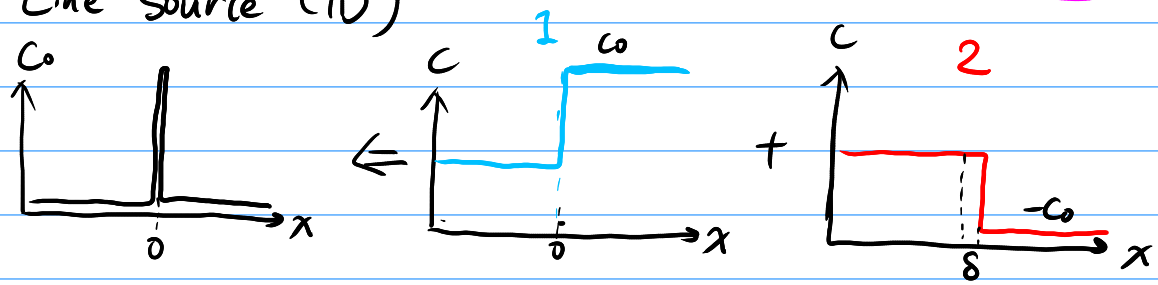
$$C(x,t) = \bar{c} + \frac{\Delta C}{2}\,\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right)$$



what if $t \to \infty$?

$$\lim_{t\to\infty} \mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) = 0 \qquad C(x,\infty) = \bar{c}$$

② Line Source (1D)



Line source is combined 2 semi-infinite sources!

Superposition $\longrightarrow$ $C(x,t) = \underbrace{C_1(x,t)}_{\text{Solution for 1}} + \underbrace{C_2(x,t)}_{\text{Solution for 2}}$

$$\left.\begin{aligned} C_1 &= \frac{C_0}{2} + \frac{C_0}{2}\,\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) \\ C_2 &= -\frac{C_0}{2} + -\frac{C_0}{2}\,\mathrm{erf}\left(\frac{x-\delta}{\sqrt{4Dt}}\right) \end{aligned}\right\} \Rightarrow C(x,t) = \frac{C_0}{2}\left[\mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right) - \mathrm{erf}\left(\frac{x-\delta}{\sqrt{4Dt}}\right)\right]$$

$$C(x,t) = \frac{C_0}{2} \cdot \frac{2}{\sqrt{\pi}} \left[\int_0^{\frac{x}{\sqrt{4Dt}}} e^{-\xi^2} d\xi - \int_0^{\frac{x-\delta}{\sqrt{4Dt}}} e^{-\xi^2} d\xi \right]$$

$$= \frac{C_0}{\cancel{2}} \frac{\cancel{2}}{\sqrt{\pi}} \boxed{\int_{\frac{x-\delta}{\sqrt{4Dt}}}^{\frac{x}{\sqrt{4Dt}}} e^{-\xi^2} d\xi} \Rightarrow \text{function} \times \text{distance}$$

total number

$$= \frac{\boxed{C_0 \delta}}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right)$$

$$= \underbrace{\frac{N}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right)}_{\text{Gaussian ! Normal distribution}}$$

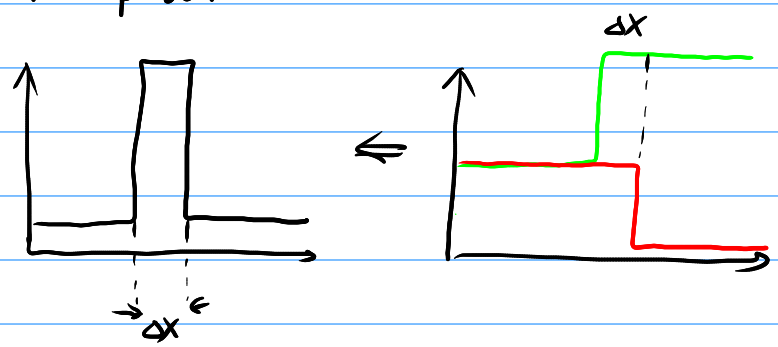
$$N(0, \sigma^2) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot \exp\left(-\frac{x^2}{2\sigma^2}\right) \qquad \sigma = \sqrt{2Dt}$$

Physical meaning : 1) Point source has $c(x,t)$ follow Gaussian

2) variance $\sigma = \sqrt{2Dt}$

3) variance increase by $t^{\frac{1}{2}}$ $\Rightarrow$ broader over time

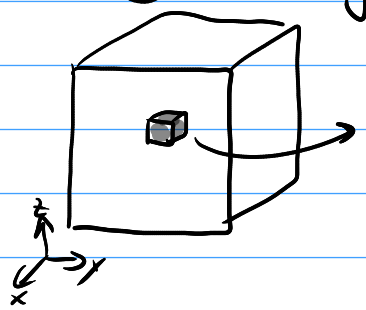
③ A step source



1) Same solution as in 2, just 2 error function superposition

2) thin film $\Delta x \to 0$ the same as point source

④ 3D geometry



Diffusion in 3-axes are uncorrelated

$$c(x,y,z,t) = \hat{c}(x,t)\cdot\hat{c}(y,t)\cdot\hat{c}(z,t)$$

$$= \frac{N}{\sqrt{(4\pi Dt)^3}}\cdot e^{-\frac{(x^2+y^2+z)^2}{4Dt}}$$

Againt the variance $\sigma = \sqrt{2Dt}$ is observed

⑤ General case

In principle, as diffusion eqn is linear PDE
we can use superposition to "fit" any initial condition

1) from semi-infinite space $\Rightarrow$ superposition of erf

2) from point / line source $\Rightarrow$ superposition of Gaussian

Solving boundary condition can be cumbersome

Method 2 Separation of variables

For diffusion in finite domain, separation of variables more often used.

We separate spatial domain with time domain

$$c(x,t) = X(x)\,T(t)$$

$$\frac{dc}{dt} = \frac{d(XT)}{dt} = X\frac{dT}{dt} + \cancel{T\frac{dX}{dt}}$$

$$\frac{d^2c}{dx^2} = \frac{d\left(\frac{d(XT)}{dx}\right)}{dx} = \frac{d\left(T\frac{dX}{dx} + \cancel{X\frac{dT}{dx}}\right)}{dx} = T\frac{d^2X}{dx^2} + \frac{dT}{dx}\cdot\frac{dX}{dx}$$

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} \Rightarrow X\frac{dT}{dt} = DT\frac{d^2X}{dx^2}$$

$$\underbrace{\frac{1}{DT}\frac{dT}{dt}}_{t\text{-only}} = \underbrace{\frac{1}{X}\frac{d^2X}{dx}}_{x\text{-only}}$$

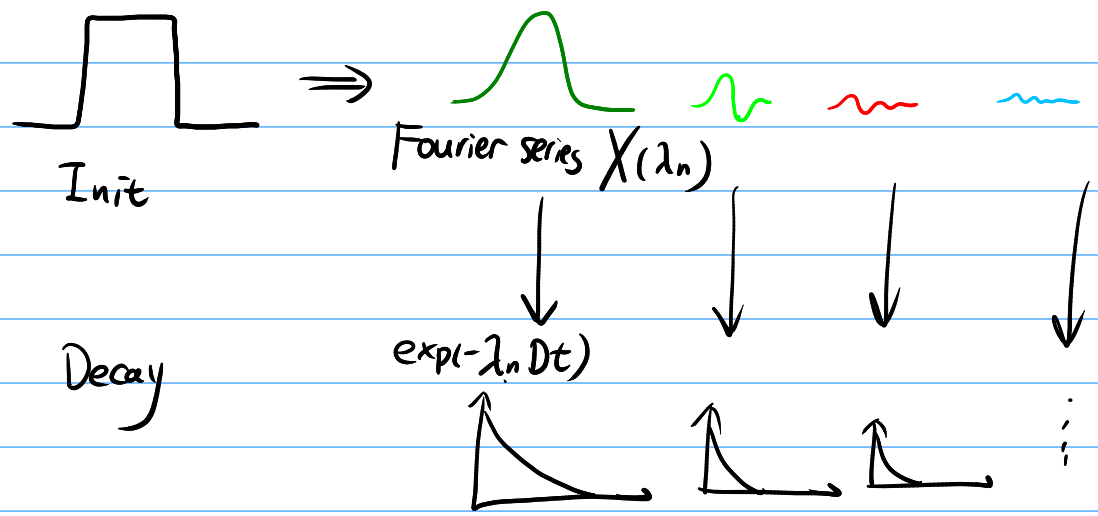
If always true LHS = RHS = const

Let const $= -\lambda^2$ (decay) (λ chosen only to make $\lambda^2 > 0$)

LHS $\frac{1}{DT}\frac{dT}{dt} = -\lambda^2 \Rightarrow \frac{dT}{dt} = -\lambda^2 DT \Rightarrow \underline{T = \exp(-\lambda^2 Dt)}$ Decay

RHS $\frac{1}{X}\frac{d^2X}{dx^2} = -\lambda^2 \Rightarrow \underbrace{\frac{d^2X}{dx^2} = -\lambda^2 X \Rightarrow X = Ae^{i\lambda x} + Be^{-i\lambda x}}_{\text{wave!}}$

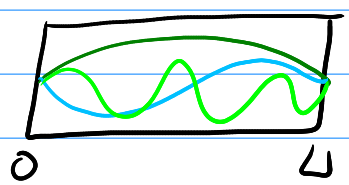
1) There are infinite λ that satisfies the equ.
2) Think diffusion eqn now can by solved by series of $\{(A_1, B_1, \lambda_1), (A_2, B_2, \lambda_2) \cdots\}$ that satisfy $\begin{cases} \text{init} \\ \text{BC} \end{cases}$



$$C(x,t) = \sum_{i=0}^{\infty} X_i(\lambda_i)\, T(\lambda_i)$$

What is the meaning of λ_i?

Spatial wave length! $\lambda \propto \frac{n}{L}$ ← (series order ; Quantum!)



Method 3 Laplace transform

We see from separation of variables,

Spatial $\sim e^{\pm i\lambda x}$

time $\sim e^{-\lambda D t}$

we have the same general solution

$\Rightarrow$ solve only coefficients

Laplace transform focuses on trying out the time ($e^{-pt} \Leftarrow$ trial func)

$$C = f(x,t) \qquad\qquad \hat{C} = F(x,p)$$

$$(x,t) \qquad\qquad (x,p)$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\boxed{f} \xleftrightarrow[\text{(Laplace transform)}]{\mathcal{L}} \boxed{F}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$C \qquad \mathcal{L} = \int_0^\infty e^{-pt} f \, dt \qquad \hat{C}$$

Laplace transform gets rid of t by another variable p

why useful ?

function $f(x,t) \xrightarrow{\mathcal{L}} \hat{f}(x,p)$

$$\hat{C}(x,p) = \hat{f}(x,p) = \int_0^\infty e^{-pt} f(x,t)$$

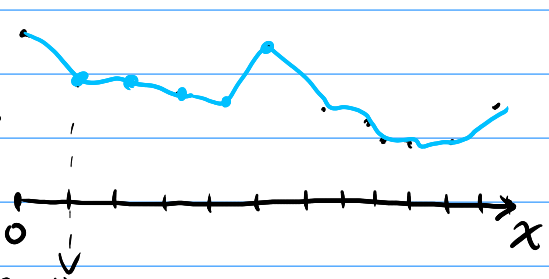
(compare fourier Transform $F(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi\xi x} dx$) just the transformation on $t \to p$ domain

B.C

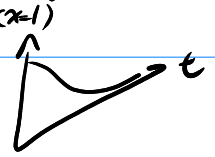
$C = C_0$

or

$\frac{\partial C}{\partial x} = J_0$



$C(x=1)$



$\Leftarrow$ Question for Laplace transform

How do C at certain x decay ?

The power of Laplace transform for initial & BC problems

$$\mathcal{L}\left(\frac{\partial f}{\partial t}\right) = \int_0^\infty \underbrace{e^{-pt}}_{u} \underbrace{\frac{\partial f}{\partial t}}_{dv/dt} dt$$

integral by parts ↓

$$= uv\Big|_0^\infty - \int v \boxed{\frac{du}{dt}} dt \quad \Rightarrow -p\frac{du}{dt}$$

$$= e^{-pt} f(x,t)\Big|_{t=0}^{t=\infty} + p \underbrace{\int_0^\infty e^{-pt} f(x,t)\, dt}_{\mathcal{L}(f)}$$

$f(x, t=\infty) = 0$

∴ time-derivative Laplace transform

$$\mathcal{L}\left(\frac{\partial f}{\partial t}\right) = p\,\mathcal{L}(f) - e^{-p(t=0)} \cdot f(x, t=0)$$

$$= p \cdot \mathcal{L}(f) - \underline{\underline{f(x, t=0)}}$$

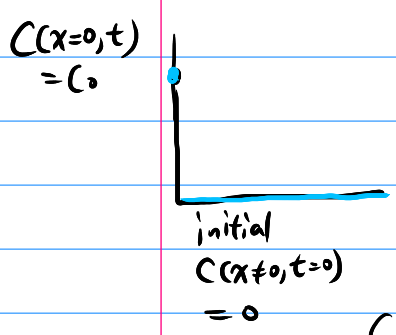
Initial values of f !

On spatial domain (Laplace transform on t, not x)

$$\mathcal{L}\left(\frac{\partial^n f(x,t)}{\partial x^n}\right) = \frac{\partial^n \mathcal{L}(f)}{\partial x^n} = \frac{\partial^n \hat{f}(x,p)}{\partial x^n}$$

Fick's 2nd law in Laplace transform

$$\mathcal{L}\left(\frac{\partial c}{\partial t}\right) = \mathcal{L}\left(D\frac{\partial^2 c}{\partial x^2}\right) \qquad c(x,t) \xrightarrow{\mathcal{L}} \hat{c}(x,p)$$



$$p \cdot \hat{c}(x,p) - c(x, t=0) = D\frac{\partial^2 \hat{c}}{\partial x^2}$$

Solving sample semi-infinite with B.C

$c(x, t=0) = 0$

$$\frac{\partial^2 \hat{c}}{\partial x^2} = \frac{p}{D}\hat{c}(x,p)$$

$$\frac{\partial^2 \hat{c}}{\partial x^2} = \frac{p}{D}\hat{c}(x,p)$$

1) Only ODE for x, not p!
2) Initial values already included

$\hat{c}(x,p)$ has general solution

$$\hat{c}(x,p) = A\exp\left(\sqrt{\frac{p}{D}}\,x\right) + B\exp\left(-\sqrt{\frac{p}{D}}\,x\right)$$

How to solve A & B?

at $x=0$ $\mathcal{L}(c(x=0,t)) = \mathcal{L}(c_0)$

$$\hat{c}(x=0,p) = \int_0^\infty e^{-pt} c_0\, dt$$

$$= -\frac{c_0}{p} e^{-pt}\Big|_0^\infty = \frac{c_0}{p}$$

Check Laplace transform table

$x=\infty$ $\mathcal{L}\left(\frac{\partial c}{\partial x}\right) = 0$

x not t

$$\frac{\partial \hat{c}(x,p)}{\partial x}\Big|_{x=\infty} = 0$$

We have $x=0$ $\hat{c}(x=0,p) = A + B = \frac{c_0}{p}$

$x=\infty$ $\frac{\partial \hat{c}}{\partial x} = 0$ A must be 0

$$\therefore B = \frac{c_0}{p} \Rightarrow \underline{\hat{c}(x,p) = \frac{c_0}{p}\exp\left(-\sqrt{\frac{p}{D}}\,x\right)}$$

Go back to table

$\hat{c}$ $\frac{e^{-qx}}{p}$ form

$\Updownarrow$

c erfc form

$$c(x,p) = c_0 \cdot \mathrm{erfc}\left(\frac{x}{\sqrt{4Dt}}\right)$$

$$= c_0\left(1 - \mathrm{erf}\left(\frac{x}{\sqrt{4Dt}}\right)\right)$$

OR, integral form $c(x,t) = \int_0^\infty e^{pt}\hat{c}(x,p)\,dp$

(less used than fourier) if p is real

General solution for Laplace transform

① Diffusion eq.

$$\frac{\partial^2 \hat{c}}{\partial x^2} - \frac{p}{D}\hat{c} = -\boxed{\frac{C(x,t=0)}{D}} \Rightarrow \text{initial}$$

② Solve above ODE to analytical form $\xrightarrow{\text{No}}$ Use numerical ?

$\downarrow$ Yes

$$\hat{c} = \hat{f}(x, p; \underbrace{A_1, A_2, A_3 \cdots}_{\text{coefficients (usually 2)}})$$

③ Fit B.C to get $A_1, A_2 \cdots$

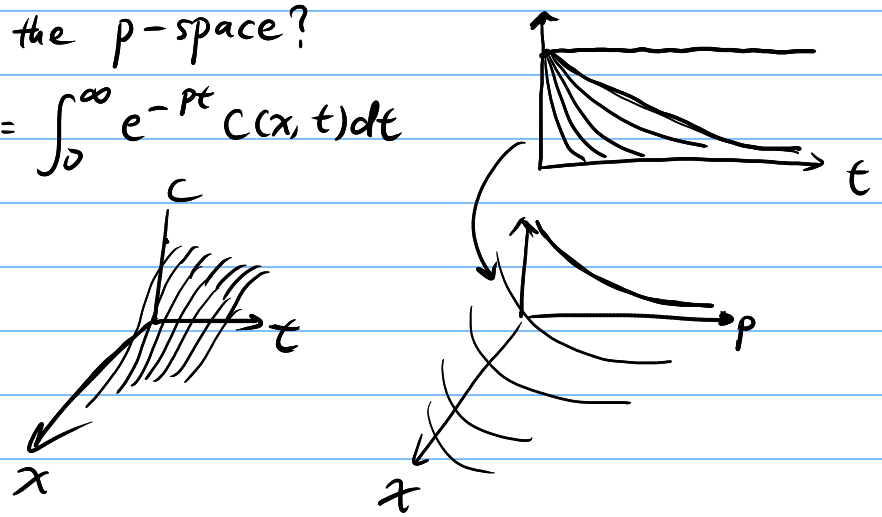
Dirichlet $\hat{c}(x = x_0, p; A_1 A_2 \cdots A_3) \Rightarrow$ Some form with $p, A_1, A_2 \cdots$

Newman $\frac{\partial \hat{c}}{\partial x}(x = x_0, p; \cdots) \Rightarrow$ Some form with $p, A_1 \cdots$

④ Get analytical $\hat{c}(x,p)$ $\xrightarrow{\text{not in table}}$ Numerical inversion

$\downarrow$ In table

Analytical solution to $c(x,t)$

Meaning of the p-space?

$$\hat{c}(x,p) = \int_0^\infty e^{-pt} c(x,t)\,dt$$